INCH-POUND

GG-L-260E
December 23,1988
SUPERSEDING
GG-L-260D
December 30, 1985

FEDERAL SPECIFICATION

LEVELS, SURVEYING: ABNEY TYPE

This specification was approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal Agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two types of Abney surveying levels.

1.2 Classification. Levels are of the following types, as specified (see 6.2):

Type I - With glass reflector.
Type II - With metal reflector.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Federal Standards:

FED-STD-H28 - Screw-Thread Standards For Federal Services.
FED-STD-123 - Marking for Shipment (Civil Agencies).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6675

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(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specification:

MIL-S-3534	- Surveying, Instruments and Accessories, Packaging of.

Military Standards:

MIL-STD-105	- Sampling Procedures and Tables For Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-889	- Dissimilar Metals.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN; NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those in effect on the date of the solicitation.

Federal Regulations:

Code of Federal Regulations (CFR)

41 CFR 1-1.25	- Recovered Materials.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

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3. REQUIREMENTS

3.1 Description. The surveying levels, hereinafter referred to as "level", shall be an Abney-type surveying instrument capable of measuring slope in not less than three scales. The level shall be furnished in a carrying case.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The level shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the level may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.4 Metal. Metal parts shall be brass or aluminum, except that the interchangeable arcs shall be aluminum, corrosion-resistant steel, or nickel silver. The reflector of the type II level shall have a polished chromium plating.

3.3.5 Glass. Glass used in the half lens shall be of optical quality, free of imperfections that distort or impair the field of view or focus. The reflector of the type I level shall be of transparent float, plate or sheet glass of select mirror quality.

3.3.6 Fabric. Fabric used in the case shall be leather, artificial leather, natural or plastic. The fabric shall not deposit any of its finish on the level.

3.3.7 Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 (41 CFR 1-1.25) to the maximum extent practical.

3.4 Design. The level shall consist of a sighting tube, an arc plate, an index arm, a bubble vial and two interchangeable, graduated scale arcs. The level shall have a nonslip mechanical or friction drive to adjust the index arm and bubble vial simultaneously, and a knurled knob locking mechanism to hold

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their position after adjustment. The level shall have a flat bottom surface, which will support the level in an upright position, and a stud or ring for attachment of a neck strap.

3.4.1 Focusing. Focusing may be internal, with external control, or by draw tube slide.

3.4.2 Arc interchange. The means of holding the arcs in position on the arc plate shall assure proper arc alignment and shall permit manual interchange or reversal of the arcs.

3.5 Construction. The attachment of parts to the level which are nonremovable shall be by soldering, brazing or other equally strong attaching method. Removable parts shall be attached by threaded fasteners.

3.5.1 Screw threads. All screws and screw threads in the level shall be in accordance with FED-STD-H28, class 2 fit.

3.5.2 Bubble vial liquid. The liquid in the bubble vial shall be an anti-freeze solution.

3.5.3 Arc mounting. The interchangeable arcs, when mounted together on the arc plate, shall be flush with the pointer on the index arm.

3.5.4 Alignment and calibration. All alignment and calibration adjustments shall be made before delivery.

3.5.5 Carrying case. The carrying case shall be a close fitting container, with a belt loop. The opening for the belt loop shall accommodate a 2-inch web belt. The case closure flap shall have a fastener to keep the case closed. If a bubble vial adjusting tool (see 3.5) is furnished with the level, the carrying case shall have a provision to securely fasten the tool inside the case.

3.6 Maintainability. The sighting tube shall be capable of disassembly to the extent that the interior can be cleaned. The reflector shall be sleeve or plate mounted for removal. If the bubble vial is mounted in an adjustable bracket, an adjusting tool shall be furnished with the level.

3.7 Performance. The level shall provide the user with an indication of the slope angle, above and below the horizontal, between the level's position and the target object's position, by observation of the index arm's pointer on an arc's graduated scale, after the user has aligned the sighting tube's internal sighting reference on the target object on one side of the sighting tube's field of view with the reflected bubble image which appears on the other side of the field of view.

3.7.1 Sighting picture. The user's sighting picture through the sighting tube shall encompass the full field of view of the sighting tube. When focused, the sighting reference and bubble image shall be clear and sharply defined.

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3.7.2 Accuracy. The total level error shall be not more than 0.50 degree, or its equivalent on either arc, through 60.0 degrees or equivalent on either arc, above and below the horizontal.

3.7.3 Bubble vial sensitivity. The bubble vial of the level shall have a sensitivity of not less than 2 millimeters (mm) of movement for 20 minutes of change or arc in the index arm.

3.8 Detailed requirements.

3.8.1 Graduated scales of arcs. One side of one arc shall have a graduated degree scale and the other side shall have a graduated percent scale. One side of the other arc shall have a graduated topographic scale and the other side shall have a graduated chain correction scale, or may be blank. The graduated scales shall be in accordance with table I.

TABLE I. Scale graduations.

Scale	Range	Lines	Numbers
Degree	60 degrees above or below horizontal	Each degree. Fifth degree line longer.	Each 10 degrees.
Percent	150 percent above or below horizontal	Each percent, 0 to 60. Fifth percent line longer. Each 2 percent, 60 to 120. Ten percent line longer. Each 5 percent, 120 to 150. Ten percent line longer.	Each 10 percent, 0 to 60. Each 20 percent, 60 to 120. At 150 percent.
Topographic	100 above or below horizontal	Each unit, 1 to 45. Fifth unit line longer. Each 5 units, 45 to 70. Each 10 units, 70 to 100.	Each 10 units, 0 to 80. At 100.

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TABLE I. Scale graduations. (Cont'd)

Scale	Range	Lines	Numbers
Chainage Correction	0 to 25 above or below horizontal	Each 1/10 unit, 0 to 1. 0.5 unit longer. Each half unit, 2 to 15. Each unit, 15 2 to 25.	First unit. Each 2 units, 2 to 8. Each 5 units, 10 to 25

3.8.2 Type I level. The type I level shall have a glass prism reflector (see 6.2).

3.8.3 Type II level. The type II level shall have a polished chromium plated reflector (see 6.2).

3.9 Dimensions. The length of the sighting tube shall be not less than 6 inches (152.4 mm) and not more than 8 inches (203.2 mm). The width of the largest cross section area of the sighting tube's exterior shall be not less than 0.65 inches (16.5 mm) nor more than 0.75 inches (19.0 mm). The diameter of the eyepiece shall be not greater than the width of the largest cross section area of the sighting tube.

3.10 Color. The interior of the sighting tube shall be a flat, nonreflective black. The scale graduations on the arc shall provide a visible contrast with the arc background. Unless otherwise specified (see 6.2), the other portions of the level that are normally painted, shall be cleaned, treated and painted in accordance with the manufacturer's standard practice.

3.11 Finish. The eyepiece shall have rounded edges on any surface that will contact the user.

3.12 Identification marking. The manufacturer's name or trademark shall be stamped or otherwise permanently affixed on the level.

3.13 Workmanship. All exposed edges of the level shall be rounded or beveled. All threaded fasteners mounting components, or holding components in alignment or calibrated position, shall be secured to prevent movement of the fastener. Moveable parts shall move smoothly and positively throughout the entire range of movement. The level shall be clean and free from any extraneous material. The carrying case shall have no gaps in any seams and shall be clean.

4. QUALITY ASSURANCE PROVISIONS.

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4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.3).
c. Inspection for preparation of delivery (see 4.5).

4.3 First article inspection.

4.3.1 Examination. The first article shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 1.0 percent defective for major defective and 4.0 percent defective for minor defects.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2. AQL shall be 1.0 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The levels shall be examined for the following defects:

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Defect No.	Defect	Requirement paragraph
	Major	
101.	Materials not as specified.	3.3
102.	Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration of the applicable storage and operating environments.	3.3.1
103.	Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.	3.3.2
104.	Contractor does not have documentation available for identification of materials, material finishes or treatment.	3.3.3
105.	Metal parts not as specified.	3.3.4
106.	Glass not as specified.	3.3.5
107.	Fabric not as specified.	3.3.6
108.	Design of level not as specified.	3.4
109.	Focusing not as specified.	3.4.1
110.	Arc interchange method not as specified.	3.4.2
111.	Construction not as specified.	3.5
112.	Bubble vial liquid not as specified.	3.5.2
113.	Arc mounting not as specified.	3.5.3
114.	Level not aligned and calibrated.	3.5.4
115.	Carrying case not as specified.	3.5.5
116.	Maintainability of level not as specified.	3.6
117.	Performance not as specified.	3.7
118.	Sight picture not as specified.	3.7.1
119.	Graduated scales on arcs not as specified.	3.8.1
120.	Type I level not as specified.	3.8.2
121.	Type II level not as specified.	3.8.3
122.	Dimensions not as specified.	3.9
123.	Color not as specified.	3.10
124.	Finish not as specified.	3.11
	Minor	
201.	Screws or screw threads not as specified.	3.5.1
202.	Identification marking not as specified.	3.12
203.	Workmanship not as specified.	3.13

4.5.2 Tests.

4.5.2.1 Accuracy test. Fasten the level to the surface of a fixture that can be inclined both up and down through 60 degrees of arc. Use an accurate protractor to determine inclinations of the fixture. First, level the fixture with an accurate bubble level in a true horizontal plane. Sight through the sighting tube of the test level, focus, and align the sighting reference on one side of the sight picture with the center of the bubble image on the other side of the sight picture by adjusting the index arm. When the sighting reference and bubble image are aligned, observe the position of the pointer of the index arm in

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relation to the graduated scale on the arc. Repeat the procedure with the fixture inclined both upward and downward at 15 degrees, 45 degrees and 60 degrees from the horizontal, carefully aligning the sighting reference and center of the bubble image at each inclined position, and observe the relationship of the pointer to the arc scale. Any evidence of error in the relationship of the pointer to the arc scale that exceeds 0.50 degree or its equivalent on either arc shall constitute failure of this test (see 3.7.2).

4.5.2.2 Vial sensitivity test. The bubble vial shall be mounted on a smooth horizontal beam. One end of the beam shall be supported on a pivot and the other by a micrometer screw. The beam shall be calibrated so that each revolution of the micrometer screw shall result in a known angular displacement of the beam, both above and below the horizontal plane, to within one minute of arc. The bubble movement within the vial shall be measured with a metric scale for movements of the beam both above and below the horizontal plane. A bubble movement that is less than 2 mm in the vial for each 20 minutes of arc movement of the beam in either direction from the horizontal shall constitute failure of this test (see 3.7.3). Unless otherwise specified (see 6.2), the contractor may certify compliance in lieu of testing.

4.5.2.3 Failure criteria. Nonconformance to 3.6 shall constitute failure of these tests.

4.6 Inspection of preparation for delivery. The preservation, packaging and packing for levels A and B shall be examined to determine compliance with the quality assurance provisions of MIL-S-3534. Inspection of commercial preservation, packaging and packing, and the inspection of marking for military and civil agencies shall be as follows:

4.6.1 Unit of product. For the purpose of examination, a completed pack, ready for shipment, shall be considered a unit of product.

4.6.2 Sampling. Sampling for examination of commercial preservation, packaging and packing, and sampling for examination of marking for military and civil agencies, shall be in accordance with MIL-STD-105.

4.6.3 Examination. Samples selected in accordance with 4.5.2 shall be examined to determine compliance with the applicable requirements of 5.1.3, 5.2.3 and 5.3. The AQL shall be 1.0 percent defective.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A, level B, or commercial, as specified (see 6.2).

5.1.1 Level A. Each level shall be preserved and packaged in accordance with the level A preservation and packaging requirements of MIL-S-3534.

5.1.2 Level B. Each level shall be preserved and packaged in accordance with the level B preservation and packaging requirements of MIL-S-3534.

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5.1.3 Commercial. The levels shall be preserved and packaged in accordance with normal commercial practice. The complete package shall be designed to protect the levels against damage during multiple shipments, handling and storage.

5.2 Packing. Packing shall be level A, level B, or commercial, as specified (see 6.2).

5.2.1 Level A. Levels, preserved and packaged as specified in 5.1, shall be packed in accordance with the level A packing requirements of MIL-STD-3534.

5.2.2 Level B. Levels, preserved and packaged as specified in 5.1, shall be packed in accordance with the level B packing requirements of MIL-S-3534.

5.2.3 Commercial. The levels, preserved and packaged as specified in 5.1, shall be packed in fiberboard boxes. The complete pack shall be designed to protect the levels against damage during multiple shipments, handling and storage, and shall assure common carrier acceptance and safe delivery, at lowest ratings, in compliance with carrier rules applicable to the mode of shipment.

5.3 Marking.

5.3.1 Military agencies. In addition to any special or identification markings required by the contract or purchase order, all containers shall be marked in accordance with MIL-STD-129.

5.3.2 Civil agencies. Marking shall be in accordance with FED-STD-123.

6. NOTES

6.1 Intended use. The level is used for measuring land slope in degrees or percent of grade and for topographic surveying.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

a. Title, number and date of this specification.
b. Time frame required for submission of first article and number of units required (see 3.2).
c. Type of level required (see 1.2, 3.8.2 and 3.8.3).
d. When treatment and painting other than manufacturer's standard practice is required (see 3.10).
e. When a certificate of compliance for vial sensitivity is not acceptable (see 4.4.2.2).
f. Degree of preservation and packaging required (see 5.1) and degree of packing required (see 5.2).
g. Any special marking (see 5.3.1).

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6.3 First article. When a first article inspection is required, the item(s) should be a first article sample. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Subject term (key word) listing.

Abney type
Level
Measuring slope
Surveying

6.5 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

MILITARY INTERESTS:

Custodians:
Army - ME
Navy - YD

Review activity:
DLA - GS

User activities:
Army - CE
Navy - MC

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - FSS
USDA - AFS
HUD - HCC

Preparing activity:
Army - ME

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
GG-L-260E	Levels, Surveying: Abney Type

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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